Finiteness of Shallow Logical Operators and a Sharp Depth Threshold

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For a quantum code on n finite-dimensional physical subsystems, we consider all exact logical unitary circuits of depth at most $h \geq 1$, with gates on at most q subsystems. For $q \geq 2$ and code distance $d \geq 2$, these circuits have only finitely many logical actions, up to phase, whenever $h < 2\lceil \log_q d \rceil - 1$, where no stabilizer or geometric locality assumption is needed, and a gauge subsystem is allowed. In contrast, for every qubit stabilizer code encoding at least one qubit, an infinite logical family is available at depth $2\lceil \log_q d \rceil - 1$, for $q \geq 2$ and $d \geq 2$. This is the exact critical depth for the onset of infinitely many logical actions in that class.

Transversal gates limit the spread of errors, but their logical actions are restricted by error detection [1–3]. Eastin and Knill proved finiteness, up to phase, for product unitaries on arbitrary finite-dimensional subspace codes whose individual parts are error-detecting [3]. For such codes, their result already gives the depth-one case considered here: apply it to each partition into correctable blocks and take the finite union over all partitions. For greater depth, successive layers can have intersecting supports. Circuits subject to a common depth bound need not form a group, so the compact Lie-group argument does not apply directly.

Here we apply the same infinitesimal restriction to circuit parameters. Detectable sums of generators on overlapping erasable regions have also been studied in the theory of covariant error correction [4, 5]. Polynomial circuit maps, their conjugated gate-insertion derivatives, and semialgebraic stratification are established tools in quantum circuit analysis [6]. The required code condition is error detection, for either subspace or subsystem codes [7, 8].

Earlier Clifford-hierarchy bounds use geometric or algebraic assumptions beyond gate arity and distance. Bravyi and König assume local stabilizer checks and geometrically local gates [9]. Pastawski and Yoshida extend the approach to Pauli subsystem codes [10]; their geometry-free Lemma 5 assumes transversality across cleanable regions. In D spatial dimensions, their topological Corollary 1 gives level $D+1$ of the Clifford hierarchy. Their Theorem 4 gives level D when the loss threshold is positive and $d = \Omega((\log n)^{1-1/D})$. Jochym-O'Connor, Kubica, and Yoder permit geometrically nonlocal gates, but their disjointness inequality need not hold at any hierarchy level when gates have arity greater than one [11].

Finiteness also occurs outside the Pauli setting. Beverland et al. prove it for protected-gate equivalence classes in two-dimensional topological field theories [12]. Their equivalence relation may remove separate phases on invariant sectors, whereas here only a single global phase is removed. The bounded-spread theorem of Webster et al. instead restricts universal gate sets on infinite-disjointness stabilizer-code families [13]; that family assumption is not made here. Our proof uses a finite smooth decomposition of a polynomial constraint set [14].

The upper-bound construction uses established Pauli-rotation synthesis. Binary-tree compression appears in Ref. [15], Sec. IV and Fig. 2(a). Cowtan et al. give balanced-tree phase gadgets with logarithmic CNOT depth [16, Cor. 3.4 and the following discussion], and Haah, Liu, and Tan explicitly obtain logarithmic-depth Pauli rotations with all-to-all connectivity [17, Cor. 3.6]. Choosing a weight-d logical Pauli gives the logarithmic upper bound for stabilizer subspace codes as a direct application. The depth-one case with gate arity at least d is also noted explicitly by Dua et al. [18, Sec. I], using a minimum-weight logical Pauli as a rotation generator.

Syndrome-resolved protocols realize continuous logical rotations through measurement and outcome-dependent recovery [19]. Constant quantum-depth injection also allows measurements and classical processing [20]. Such protocols lie outside the exact unitary circuit model considered here. Bounds on fault-tolerance overhead concern a different resource. Physical-width and cumulative spacetime bounds depend on storage duration, noise, and target error [21, 22]; nearly logarithmic- and logarithmic-time compilers concern noisy simulation of a finite source alphabet [23, 24].

Theorem 1 gives a distance criterion for finiteness under one common total-depth bound, without assigning a Clifford-hierarchy level. For stabilizer subspace codes, Theorem 2 identifies the first depth at which the logical image is infinite. Figure 1 illustrates the two mechanisms.

Let the physical system be

$$\mathcal{H} = \bigotimes_{i=1}^{n} \mathcal{H}_i,$$

with each $\mathcal{H}_i$ finite-dimensional. An isometry

$$V : \mathcal{H}_L \otimes \mathcal{H}_G \longrightarrow \mathcal{H}, \qquad \dim \mathcal{H}_L = K \geq 2,$$

specifies the code. We assume $g = \dim \mathcal{H}_G \geq 1$. The

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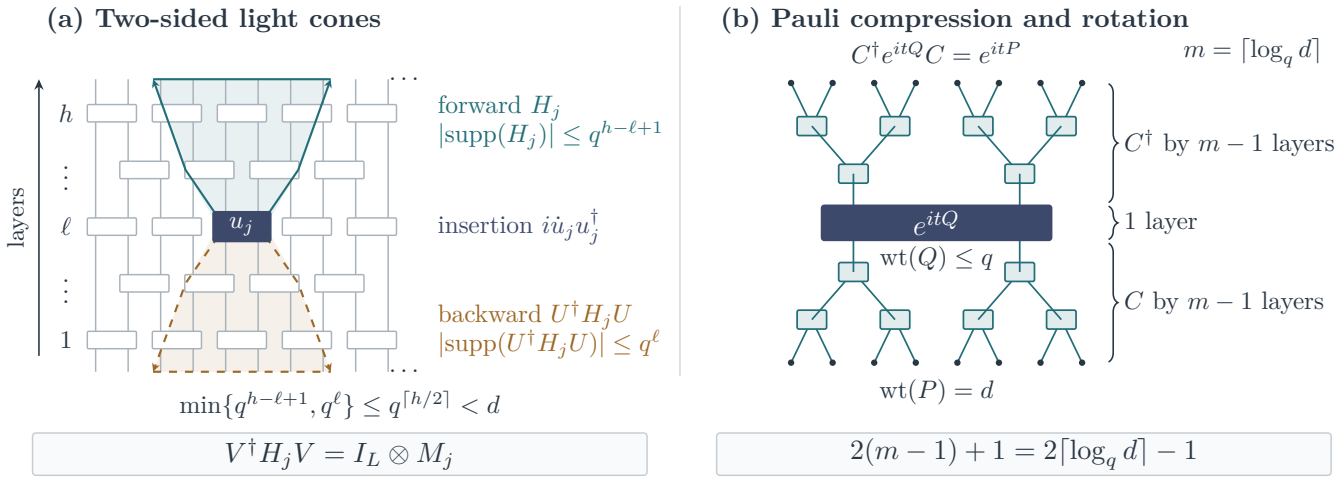


FIG. 1. Mechanisms behind Theorems 1 and 2. Time runs upward in both panels. (a) An insertion at gate u_j in layer ℓ has forward and backward supports bounded by $q^{h-\ell+1}$ and q^ℓ. When $q^{\lceil h/2 \rceil} < d$, at least one support is detectable. The exact encoded action transfers its gauge-only compression to $V^\dagger H_j V = I_L \otimes M_j$, for a gauge operator M_j. The brickwall arrangement is schematic and imposes no geometric locality. (b) With $m = \lceil \log_q d \rceil$ and $q \ge 2$, Clifford gates compress a weight-d logical Pauli to $Q = CPC^\dagger$ in $m-1$ layers. One gate e^{itQ} and $m-1$ inverse layers give e^{itP} at depth $2m-1$. The case $(q,d) = (2,8)$ is illustrated; in general, a gate may collect up to q factors. Only active Pauli support is shown; the physical qubits remain.

factor $\mathcal{H}_L$ carries the encoded information, and $\mathcal{H}_G$ allows for gauge degrees of freedom. A subspace code has $\dim \mathcal{H}_G = 1$. Let d be the code distance: every operator O supported on fewer than d physical subsystems satisfies

$$V^\dagger O V = I_L \otimes M_O \tag{1}$$

for some operator M_O on $\mathcal{H}_G$ [7, 8]. Theorem 1 uses only this detection condition, so a lower bound on the code distance suffices there.

A physical unitary U implements an exact logical operation if

$$UV = V(W \otimes Z) \tag{2}$$

for unitaries W on $\mathcal{H}_L$ and Z on $\mathcal{H}_G$. The equality holds for all logical and gauge states. We count the logical factor W, which is determined up to phase by U.

A circuit has depth at most h when it consists of at most h layers, each a product of gates on disjoint supports. Each elementary gate may act on any subset of at most q physical subsystems, with an arbitrary unitary matrix. Two circuits implementing Eq. (2) are logically equivalent when their factors W differ only by an overall phase.

Theorem 1. *Let $n \ge 1$, $h \ge 1$, and $q \ge 2$ be integers, and let V encode a quantum code satisfying Eq. (1) for an integer $d \ge 2$ on n physical subsystems. If*

$$h < 2\lceil \log_q d \rceil - 1, \tag{3}$$

then the unitary circuits of depth at most h, with elementary gates acting on at most q subsystems, that satisfy Eq. (2) form only finitely many logical equivalence classes. All gate supports and unitary matrices are allowed. Their logical actions are consequently not dense in $\mathrm{PU}(K)$.

Proof. Equation (3) is equivalent to $q^{\lceil h/2 \rceil} < d$.

Group all circuits by the gate supports in each layer. After adding singleton identity gates on idle sites, a layer is a partition of the n sites into blocks of size at most q. There are finitely many such partitions, and hence finitely many support arrangements for h layers. Circuits of smaller depth are included by identity padding. The unitary matrices on these supports remain arbitrary.

Choose one arrangement, denoted by α, and list its gates in layer order as $u_1, \ldots, u_J$, so that $U = u_J \cdots u_1$. Consider a smooth family of gate matrices and logical factors W, Z satisfying Eq. (2). Its Hermitian generator is

$$H = i\dot{U}U^\dagger = \sum_{j=1}^{J} U_{>j}\left(i\dot{u}_j u_j^\dagger\right)U_{>j}^\dagger, \tag{4}$$

where $U_{>j} = u_J \cdots u_{j+1}$. This is the conjugated gate-insertion derivative used in circuit parameter analysis [6]. Write the jth summand as H_j, and suppose that gate j is in layer ℓ. Its forward light cone has at most $q^{h-\ell+1}$ sites. In the opposite frame,

$$U^\dagger H_j U = U_{<j}^\dagger (i u_j^\dagger \dot{u}_j) U_{<j},$$

where $U_{<j} = u_{j-1} \cdots u_1$. This operator has a backward light cone of at most q^ℓ sites. Other gates in layer ℓ have disjoint supports and cancel from both counts. At least one of these two supports therefore has size at most

$$q^{\min\{\ell, h-\ell+1\}} \le q^{\lceil h/2 \rceil} < d.$$

If the forward support is smaller than d, Eq. (1) applies directly to H_j. If only the backward support is smaller, it gives $V^\dagger U^\dagger H_j U V = I_L \otimes N_j$. Equation (2) then implies

$$\begin{aligned} V^\dagger H_j V &= (W \otimes Z)(I_L \otimes N_j)(W^\dagger \otimes Z^\dagger) \\ &= I_L \otimes Z N_j Z^\dagger. \end{aligned}$$

Thus $V^\dagger H_j V = I_L \otimes M_j$ for every j, with all compressions taken in the same frame. Summing gives

$$V^\dagger H V = I_L \otimes M \tag{5}$$

where $M = \sum_{j=1}^{J} M_j$ is Hermitian on $\mathcal{H}_G$. This is the detectable-generator step [3, 4].

Differentiating Eq. (2), and using $U^\dagger V = V(W^\dagger \otimes Z^\dagger)$, also gives

$$\begin{aligned} V^\dagger H V &= A \otimes I_G + I_L \otimes B, \\ A = i\dot{W}W^\dagger, &\qquad B = i\dot{Z}Z^\dagger. \end{aligned} \tag{6}$$

Taking the partial trace over $\mathcal{H}_G$ yields

$$A = \frac{\mathrm{Tr}(M - B)}{g}\, I_L.$$

Thus $\dot{W} = -i\lambda W$ for a real scalar λ, and the logical motion is only a phase.

To obtain finiteness, let D_j be the Hilbert-space dimension on the support of gate j, and put

$$\mathcal{P}_\alpha = \left(\prod_{j=1}^{J} \mathrm{U}(D_j) \right) \times \mathrm{U}(K) \times \mathrm{U}(g).$$

Let $\mathcal{X}_\alpha \subseteq \mathcal{P}_\alpha$ consist of the tuples $(\mathbf{u}, W, Z)$ satisfying Eq. (2). Unitarity and Eq. (2) are polynomial equations in the real and imaginary matrix entries, so $\mathcal{X}_\alpha$ is a real algebraic set.

Lemma 1 in the End Matter gives a finite smooth decomposition

$$\mathcal{X}_\alpha = \bigsqcup_{r=1}^{N_\alpha} C_r, \qquad N_\alpha < \infty, \tag{7}$$

where each cell is smoothly diffeomorphic to an open cube $(0,1)^{\nu_r}$, including a point when $\nu_r = 0$. This partition also covers singular points of $\mathcal{X}_\alpha$.

For two tuples $x, y \in C_r$, choose a smooth diffeomorphism $\varphi_r : (0,1)^{\nu_r} \to C_r$, and set $a = \varphi_r^{-1}(x)$, $b = \varphi_r^{-1}(y)$. The path

$$\gamma(t) = \varphi_r\big((1-t)a + tb\big), \qquad 0 \le t \le 1, \tag{8}$$

stays in C_r. Its coordinates give smooth gate matrices and logical factors satisfying Eq. (2) throughout. The preceding calculation makes $[W]$ constant along γ. Each cell therefore contributes at most one logical operation modulo phase, giving at most N_α operations for this support arrangement.

The finitely many arrangements cover every circuit of depth at most h and gate size at most q. Their logical operations form a finite union of finite sets. Since $K \ge 2$, that union is closed and cannot be dense in $\mathrm{PU}(K)$. □

The finiteness condition has an exact boundary for stabilizer subspace codes. To state the contrast, let $\mathcal{L}_{\le h}(\mathcal{C}; q)$ be the logical actions modulo phase of all exact circuits of depth at most h, with gate arity at most q, on a code $\mathcal{C}$. For an integer $q \ge 2$, define the critical depth

$$h_\infty(\mathcal{C}; q) = \min\{h \in \mathbb{Z}_{\ge 1} : \mathcal{L}_{\le h}(\mathcal{C}; q) \text{ is infinite}\}, \tag{9}$$

with value ∞ if the set is empty. For $d \ge 2$, this definition and Theorem 1 give

$$h_\infty(\mathcal{C}; q) \ge 2\lceil \log_q d \rceil - 1. \tag{10}$$

Theorem 2. *Let $\mathcal{C}$ be an $[[n, k, d]]$ qubit stabilizer subspace code, with $k \ge 1$ and actual distance $d \ge 2$. Let $q \ge 2$ be an integer, and allow arbitrary unitary gates on any set of at most q qubits. Then*

$$h_\infty(\mathcal{C}; q) = 2\lceil \log_q d \rceil - 1. \tag{11}$$

At this depth, the logical image contains a nontrivial continuous one-parameter subgroup.

Proof. The lower bound is Eq. (10). For a stabilizer subspace code, the distance d is the minimum weight of a Hermitian n-qubit Pauli operator that preserves the code space and acts nonscalarly on it. Choose such an operator P of weight d, with $P^2 = I$, and put $m = \lceil \log_q d \rceil$.

Partition the nonidentity factors of P into groups of q, except possibly one smaller group. On each group, basis changes and CNOTs conjugate the product to a Pauli on one qubit. Their composition is a single allowed Clifford gate on that group. The groups are disjoint, so these gates form one layer. If w_r is the weight after r layers, then $w_{r+1} = \lceil w_r/q \rceil$ and $w_0 = d$, so $w_r = \lceil d/q^r \rceil$. Hence $m - 1$ layers give a Clifford circuit C such that

$$Q = CPC^\dagger, \qquad \mathrm{wt}(Q) \le q.$$

Local basis changes are included in the allowed Clifford gates and require no additional layers. When $m = 1$, take $C = I$. The family

$$U(t) = C^\dagger e^{itQ} C = e^{itP} \tag{12}$$

has depth at most $2m - 1$, since its middle factor is a single allowed gate. It preserves the code and induces $e^{it\overline{P}}$, where $\overline{P}$ is a nontrivial logical Pauli. Since $\overline{P}$ has both eigenvalues ± 1, these projective actions have period π and form a nontrivial continuous U(1) subgroup. □

Below the critical depth, the entire logical image is finite. At the critical depth, it already contains a continuous family. Theorem 2 combines the Pauli-rotation construction with Theorem 1, which excludes infinitely many logical actions at every smaller positive depth. This contrast shows that the depth bound cannot be increased using only gate arity and distance. The exact count treats an arbitrary q-qubit unitary as one elementary gate; it is not a CNOT-depth count. For arbitrary codes, Eq. (10) remains a lower bound, and equality is not asserted.

The construction uses no auxiliary qubits, measurements, or individually addressable logical qubits. The presence of this subgroup does not imply universality on the full logical register. The equality in Eq. (11) concerns subspace codes; it is not asserted for subsystem codes. Code preservation at the end of the circuit also does not establish fault tolerance during its intermediate layers.

Theorem 1 concerns a single common total-depth bound. It does not restrict the group generated by the resulting finite set: products can exceed that bound. Universal logical groups generated by few-body depth-one layers on different partitions are therefore compatible with this conclusion [18]. The theorem alone imposes no fault-tolerance overhead bound. Its finiteness is for each code separately; no uniform cardinality bound in n or K is asserted.

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END MATTER

Finite smooth decomposition of polynomial zero sets

Together with the infinitesimal argument of Theorem 1, finite Whitney stratification of the semialgebraic constraint set into connected smooth strata already gives a finite logical image. To keep the argument elementary and self-contained, we give a short proof using linear algebra and the implicit function theorem, with elementary polynomial arithmetic. The finite decomposition below is the polynomial case of standard semialgebraic theory [14]. A smooth cell below means an embedded smooth submanifold that is smoothly diffeomorphic to an open cube $(0,1)^r$, with a point allowed when $r = 0$. No smoothness of the full zero set is assumed.

Lemma 1. *For any finite family of real polynomials on $\mathbb{R}^N$, there is a finite partition of $\mathbb{R}^N$ into smooth cells on each of which every polynomial has constant sign. Consequently, the common zero set of finitely many real polynomials is a finite disjoint union of smooth cells.*

Proof. We induct on N. In one variable, the finitely many real roots of the nonzero polynomials partition the line into points and open intervals. The assertion is also immediate for $N = 0$. Write the variables in the induction step as (x, t), where $x \in \mathbb{R}^{N-1}$. Discard identically zero input polynomials, and write the remaining ones as $p_1(x,t), \dots, p_a(x,t)$.

We first specify a finite family of projection polynomials in x. For every nonempty subset $I \subseteq \{1, \dots, a\}$, form $F_I = \prod_{i \in I} p_i$ and include all its coefficients in t. For every truncation

$$f(x,t) = \sum_{j=0}^{m} f_j(x) t^j$$

of F_I, with $1 \le m \le \deg_t F_I$, also include all square minors of the coefficient matrix of the linear map

$$\begin{aligned} T_f : (A, B) &\longmapsto Af + B\partial_t f, \\ \deg A \le m-1, &\qquad \deg B \le m. \end{aligned} \tag{13}$$

The target consists of polynomials of degree at most $2m-1$, so this matrix has $2m$ rows and $2m+1$ columns. Its entries and minors are polynomials in x. Zero polynomials may be discarded. There are finitely many subsets, truncations, coefficients, and minors. By induction, partition the base $\mathbb{R}^{N-1}$ into finitely many smooth cells on which these projection polynomials have constant signs.

Fix one base cell C. Each p_i is either identically zero as a polynomial in t at every $x \in C$, or has one constant degree there. This follows from the coefficient signs. If all are identically zero, the cylinder $C \times \mathbb{R}$ is already a suitable cell. Otherwise let I be the indices of the nonzero factors and set $F = F_I$. Its degree m is constant on C. If $m = 0$, the same cylinder suffices. Suppose $m \ge 1$. The truncation at this degree agrees with F on every fiber over C, so its projected matrix represents T_F.

The projected minors make the rank of T_F constant on C. At a point x, write $F = d_0 f_0$, $\partial_t F = d_0 g_0$, where d_0 is the monic greatest common divisor and f_0, g_0 are coprime. Put $s = \deg d_0$ and $r = m - s$; then $\deg f_0 = r$, $\deg g_0 = r - 1$, and $r \ge 1$. The identity $AF + B\partial_t F = 0$ holds exactly when $A = Qg_0$ and $B = -Qf_0$. The degree restrictions in Eq. (13) give $\deg Q \le s$. Hence

$$\dim \ker T_F = s + 1, \qquad \operatorname{rank} T_F = 2m - s.$$

In particular, s and r are constant on C.

The monic squarefree part $G = f_0/c$, where c is the leading coefficient of F, has precisely the real roots of F, each simple: differentiation lowers each root multiplicity by one. We show that its coefficients depend smoothly on $x \in C$, even where F has repeated roots. Solve

$$\begin{aligned} AF + B\partial_t F = 0, &\qquad \deg A < r, \\ B = t^r + \sum_{j<r} b_j t^j. & \end{aligned} \tag{14}$$

There is exactly one solution: $A = -g_0/c$, $B = f_0/c$. The coefficient matrix for the $2r$ unknown coefficients is a column submatrix of T_F. It has full column rank: a homogeneous solution would have $\deg B < r$, whereas any nonzero kernel vector has $B = -Qf_0$ of degree at least r. Choose a nonzero $2r$-by-$2r$ minor of this coefficient matrix at one point of C. It is among the projected minors and thus remains nonzero throughout C. Cramer's rule expresses every coefficient in Eq. (14) as a rational function of x with a denominator that does not vanish on C. Thus $G = B$, the monic squarefree part, has smooth coefficients on C. The same construction for the singleton products gives a smooth monic squarefree part G_i for every nonconstant active factor p_i.

The number of real roots of G is locally constant on C. Indeed, at a point x_0, its real roots are simple. Apply the implicit function theorem in a smooth coordinate chart of C: each root persists uniquely and smoothly. A monic polynomial $t^r + \sum_{j<r} b_j t^j$ has all its real roots in $|t| \le 1 + \max_j |b_j|$. Near x_0, this gives one uniform bound. Outside small disjoint intervals about the old roots, $G(x_0, t)$ is bounded away from zero on the remaining compact interval. Continuity excludes new roots there; within the small intervals, the derivative stays nonzero and there is exactly one root. Since C is connected, the number of real roots is constant, and ordering them gives globally smooth functions

$$\theta_1(x) < \cdots < \theta_b(x).$$

Inactive factors vanish identically, and active constant factors have constant nonzero signs. For an active nonconstant factor, suppose $p_i(x_0, \theta_j(x_0)) = 0$. The simple

root of G_i at that point persists as a smooth root $\beta(x)$. Because p_i divides F, this is also a root of G, and the local uniqueness of the latter root gives $\beta(x) = \theta_j(x)$. Thus vanishing on the graph is an open condition in C. Nonvanishing is open by continuity. Connectedness makes one alternative hold everywhere, and any nonzero sign is then constant.

Partition $C \times \mathbb{R}$ into these b root graphs and the $b+1$ intervening open bands, including the unbounded bands. No active factor vanishes in a band, so its sign is constant there. A graph is diffeomorphic to C; a band is diffeomorphic to $C \times (0,1)$. For finite endpoints use affine interpolation. For a half-line use $u/(1-u)$, and for the whole line use $u/(1-u) - (1-u)/u$, where $0 < u < 1$. Thus all pieces are smooth cells. There are finitely many base cells, and each yields finitely many pieces. This completes the induction. The common zero set is the union of those cells on which all the given polynomials vanish. □

This construction includes nullified factors, degree drops, repeated roots, and singular points of the zero set. The subset products handle factors that vanish on a base cell; the truncations handle degree drops; the projected minors separate rank changes. Nothing requires the polynomial zero set itself to be a manifold.

To apply the lemma, express the unitary and encoding constraints for one circuit-support arrangement in real and imaginary matrix entries. Their common zero set is $\mathcal{X}_\alpha$. For two points in one cell, a smooth chart from an open cube carries the straight segment between their preimages to a smooth path in $\mathcal{X}_\alpha$. The circuit calculation in Theorem 1 gives $\dot{W} = -i\lambda W$ along this path, so its endpoints have the same projective logical action. Each cell contributes at most one such action. Finitely many cells, followed by finitely many support arrangements, give the claimed finiteness.